

1.18 Computer Systems, Programming Paradigms and C

methods are you using for the same? Answers to such questions will decide the body of the program. Now start writing the various steps one by one in a sequence. Thus you get a plan of the program which you may call algorithm. The Figure 1.10 illustrates the process in several steps.

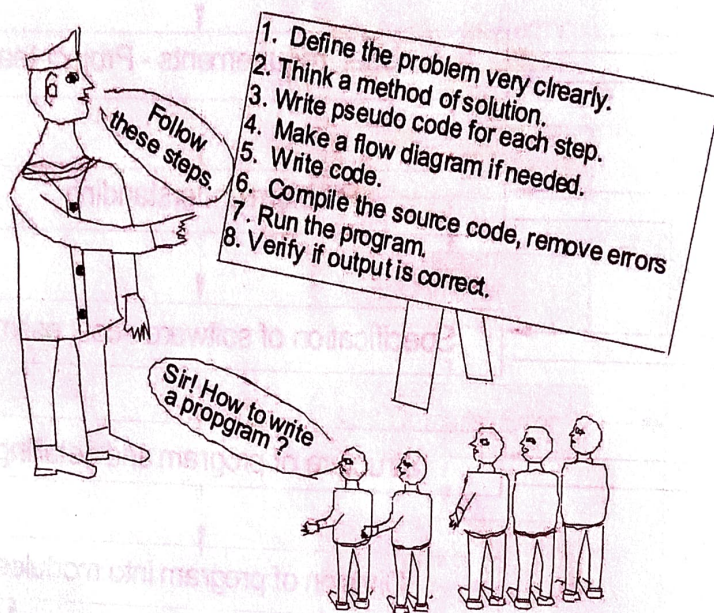


Figure 1.10 - Illustrates how to write a program

Before writing the code for a program it is helpful to make an algorithm or a flow chart of the program or write a pseudo-code.

Algorithm

An algorithm is a finite sequence of unambiguous statements about the processes involved in a program. It may be written in plain English or pseudo-code which comprises steps which when carried out lead to the solution. For example, algorithm for addition of two numbers would be:

Example 1.1 – Algorithm for adding two numbers

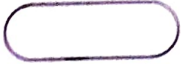
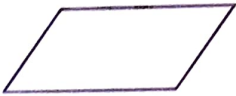
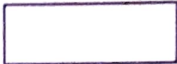
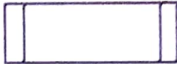
- (1) Get/ read the two numbers n_1, n_2
- (2) Define $\text{sum} = 0$
- (3) Add n_1 and n_2 to the sum, i.e. $\text{sum} = n_1 + n_2$
- (4) Display/print the sum

The name 'algorithm' is derived from the name of a 9th - century Persian mathematician Al-Khwarizmi who first wrote a book on rules for the four arithmetic operations, i.e., addition, subtraction, multiplication and division on decimal numbers. More examples of algorithms are given below.

Flow Charts

Flow charts give pictorial view of the various processes involved in the program and of the program as a whole. It gives a visual plan of the program and a chance to the programmer to review the program. The different symbols used for constructing flow charts for the various processes in a program are shown in Table 1.2.

Table 1.2 Symbols used in flow charts

Operation	Description	Symbol
(1) Start / stop	Starting or Stopping of a program	
(2) Input / Output	Process of reading data into the program / display of output on the monitor	
(3) Process	Assignment of values to variables	
(4) Predefined process	A variation of this may be used for function call.	
(5) Looping	Repeated evaluation of statements (loop)	
(6) Testing / checking	Testing whether the logical statement is true so that a decision is taken for branching.	
(7a) Connector	The symbol (circle) shows connections if part of diagram is amplified separately. Showing connections in a magnified detail.	
(7b) Connector	The symbol is used when the diagram is Spread over different pages	
(8) Arrows	The arrows connect the different process. To present a sequence of processes.	

A few examples of algorithms and flow charts are given below to illustrate how these can be useful in programming. These examples deal with simple problems.

Example 1.2 Write an algorithm and make a flow chart for writing a program which determines the area of any rectangle whose dimensions are entered by user of program.

Solution Algorithm - The algorithm is simple as given below..

- (i) Define the variables **length**, **width** and **area** and assign 0 to all.
- (ii) Read values of **length** and **width** as entered by user.
- (iii) Assign the values read to **length** and **width** respectively.

- (iv) Calculate $\text{length} \times \text{width}$ and assign the product to area.
- (v) Print or display value of area.
- (vi) End the program.

Flow Diagram: The flow diagram is shown in Figure 1.11. After start the variables length and width are defined. In next step the values 20 and 10 are read and assigned to length and width respectively. The values are then multiplied and the product is assigned to area . The arrow assignment of value.

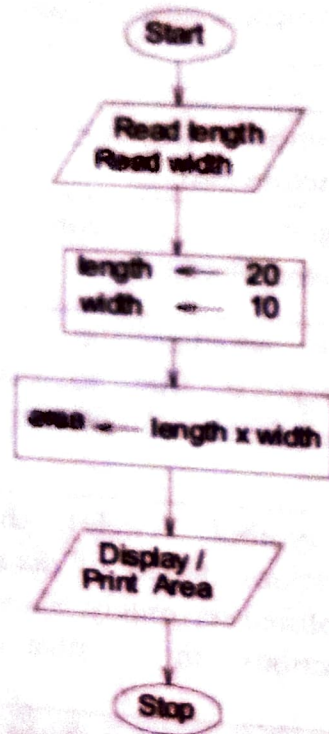


Figure 1.11 Flow diagram for the problem

Example 1.3

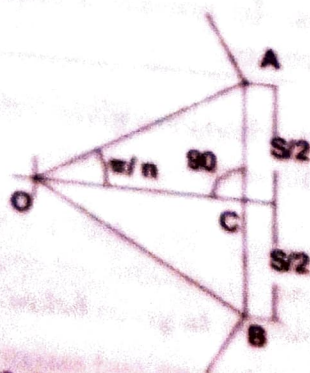


Figure 1.12 Polygon contains n such triangles

Make an algorithm and flow chart for calculation of the area of a regular polygon having n sides and each side equal to s . Values of s and n are entered by user.

Solution – Calculation of area

The polygon consists of n triangles like AOB (Fig. 1.12)

$$\text{Angle AOC} = \pi / n$$

$$\text{Area of polygon} = n \times \text{Area of triangle AOB}$$

$$= n s^2 / (4 \tan (\pi / n))$$

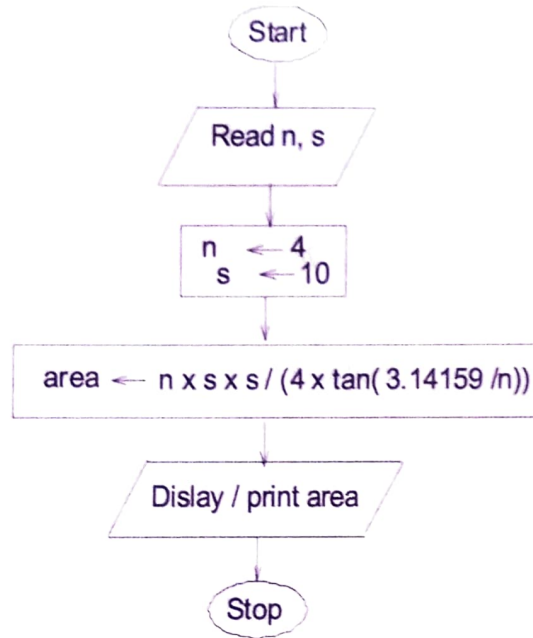


Figure 1.13 Flow chart for Example 1.3

Algorithm

- (i) Read number of sides and length of side of polygon supplied by user.
- (ii) Compute $n s^2 / (4 \tan (\pi / n))$
- (iii) Assign the result of computation to area.
- (iv) Display / print the area.
- (v) Stop the program.

The flow chart is shown in Figure 1.13. The steps involved are already discussed in the algorithm. The sequence of statements in the program is related to the 4 steps mentioned in the algorithm above.

Example 1.4 Construct algorithm and flow chart for determining the factorial of a given number n supplied by user of the program.

Solution The factorial of a number is expressed as

$$\text{Factorial } n = n! = 1 \times 2 \times 3 \times 4 \times \dots \times n.$$

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Algorithm

- (i) Declare Factorial = 1 and integer $m = 1$
- (ii) Factorial $\leftarrow$ Factorial $\times m$
- (iii) $m \leftarrow m + 1$
- (iv) If $m < n$, repeat statement (ii).
- (v) Repeat statements (iii)
- (vi) Repeat statements (ii) and (iii) till $m > n$.
- (vii) Display/print final value of Factorial
- (viii) Stop the program

The flow chart is shown below.

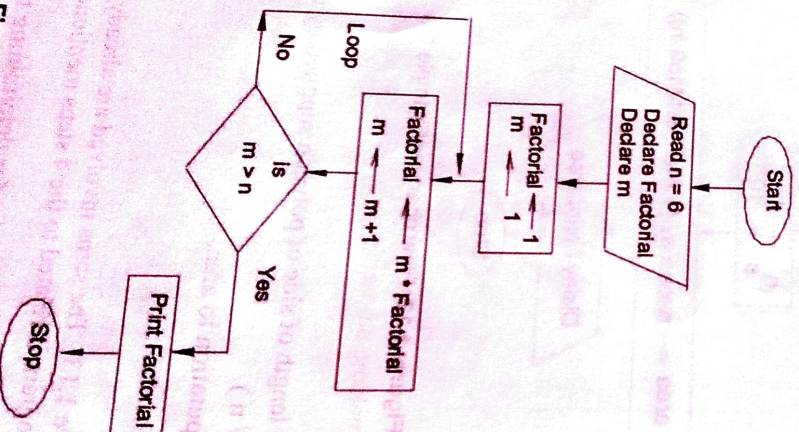


Figure 1.14 Flow chart for Example 1.4

The program involves a loop. Iteration is done m times. Value of m is increased after each iteration and factorial value is displayed/printed.

the value of m is compared with n . The next iteration is done only if $m \leq n$. When $m = n$ the loop stops.